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opheliaayamga246810-arch Update ophelia-week7-day3.md eca60ea · now

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Ophelia: Week 7, Day 3

Five Important Hyper-parameters for QLoRA

1. Target Modules

Choose which modules to target, usually Targets the attention layers first and other as well.

2. r

Number of dimensions of the LoRA matrices

3. Alpha

Multiply LoRA A and LoRA B and Alpha tell you how much to scale before adding to the biggest module.

4. Quantization

Minimizing to 8 or 4 bits, saving more space for other things. - Makes the models poorer though.

5. Dropout

Randomly Probability that get wipe / zero out during the training. Usually, 10%. It prevents your neural network from overfitting the data, improving Generalization.

Five Important Hyper-parameters for Training

1. Epoch's

Passing data through in batches.

2. Batch size

How many batch are in one size batch

3. Learning Rate

How bigger step you take each time - Not point NOT Hot one

4. Gradient Accumulation

Forward pass, backward pass, take a baby step

5. Optimizer

-

The 4 Steps of Training

Tweaking the parameters of a model based on training data, in a way that should generalize to unseen data

1. Forward pass

Predict the output given inputs

2. Loss

How different was the prediction to the ground truth

3. Backward pass

How should we tweak parameters to do better next time (the "gradients")

4. Optimization

Update parameters a tiny step to do better next time

The process of experimenting with these is called "hyper-parameter optimization"